Harder Geometrical Problems — problem 12.3.2.87

A. Engel — Problem-Solving Strategies · 1998

The circles C_1 and C_2 with centers O_1 and O_2 intersect in the points A and B. The ray O_1B intersects C_2 in F, and the ray O_2B intersects C_1 in E. The straight line through B and parallel to EF intersects the circles C_1 and C_2 a second time in M and N, respectively. Prove that $MN = AE + AF$.